Table 1: **Supplementary Table 5: Statistical comparison and model selection of constrained Gaussian mixture models on mRNA–protein fold changes.** Systematic evaluation of bivariate Gaussian mixture models across cohorts (BDL and CCl_4) and elongation factors $c \in \{1.3, 1.5, 2.0, 2.5, 4.0\}$. Models were fitted using an Expectation-Maximization (EM) algorithm to joint distributions of $N = 943$ matched gene–protein pairs. Number of free parameters P, Log-Likelihood ($\log L$), Bayesian Information Criterion (BIC), $\Delta\text{BIC} = \text{BIC} - \text{BIC}_{K=1}$, and estimated mixing proportions π_k (Diagonal π_1, Vertical π_2, Horizontal π_3). Highlighted rows denote the overall winning model for each etiology (lowest BIC).

Cohort	Model	c	Ratio $\sqrt{c}$	P	$\log L$	BIC	ΔBIC	π_1 (Diag)	π_2 (Vert)	π_3 (Horiz)	Status
Cholestatic Liver Fibrosis (BDL, $N = 943$ pairs)											
BDL	$K = 1$	–	–	5	-1830.5	3695.3	+0.0	100.0%	–	–	Null Single Gaussian
BDL	$K = 2$	1.3	1.14	10	-1661.0	**3390.5**	**-304.8**	90.9%	9.1%	–	**Winning Model (Lowest BIC)**
BDL	$K = 2$	1.5	1.22	10	-1662.3	3393.0	-302.2	91.1%	8.9%	–	Constrained Model
BDL	$K = 2$	2.0	1.41	10	-1666.8	3402.0	-293.3	91.5%	8.5%	–	Constrained Model
BDL	$K = 2$	2.5	1.58	10	-1671.5	3411.5	-283.7	92.0%	8.0%	–	Constrained Model
BDL	$K = 2$	4.0	2.00	10	-1683.5	3435.6	-259.7	92.8%	7.2%	–	Constrained Model
BDL	$K = 3$	1.3	1.14	15	-1647.9	3398.5	-296.8	85.8%	8.8%	5.4%	Constrained Model
BDL	$K = 3$	1.5	1.22	15	-1649.4	3401.5	-293.7	86.0%	8.8%	5.1%	Constrained Model
BDL	$K = 3$	2.0	1.41	15	-1653.2	3409.1	-286.2	86.5%	8.9%	4.6%	Constrained Model
BDL	$K = 3$	2.5	1.58	15	-1656.4	3415.6	-279.7	86.8%	9.0%	4.3%	Constrained Model
BDL	$K = 3$	4.0	2.00	15	-1663.4	3429.5	-265.8	87.4%	9.0%	3.7%	Constrained Model
Toxic Liver Fibrosis (CCl_4, Month 12 vs time-matched Month 12 oil, $N = 943$ pairs)											
CCl4	$K = 1$	–	–	5	-1797.5	3629.3	+0.0	100.0%	–	–	Null Single Gaussian
CCl4	$K = 2$	1.3	1.14	10	-1729.4	3527.3	-102.0	62.1%	37.9%	–	Constrained Model
CCl4	$K = 2$	1.5	1.22	10	-1729.4	3527.3	-102.0	62.1%	37.9%	–	Constrained Model
CCl4	$K = 2$	2.0	1.41	10	-1729.4	3527.3	-102.0	62.1%	37.9%	–	Constrained Model
CCl4	$K = 2$	2.5	1.58	10	-1729.4	3527.3	-102.0	62.1%	37.9%	–	Constrained Model
CCl4	$K = 2$	4.0	2.00	10	-1729.4	3527.3	-102.0	62.1%	37.9%	–	Constrained Model
CCl4	$K = 3$	1.3	1.14	15	-1692.4	**3487.6**	**-141.7**	63.6%	28.0%	8.5%	**Winning Model (Lowest BIC)**
CCl4	$K = 3$	1.5	1.22	15	-1694.9	3492.5	-136.8	64.7%	28.9%	6.4%	Constrained Model
CCl4	$K = 3$	2.0	1.41	15	-1698.8	3500.3	-129.0	66.6%	28.8%	4.6%	Constrained Model
CCl4	$K = 3$	2.5	1.58	15	-1701.5	3505.8	-123.5	67.9%	28.3%	3.8%	Constrained Model
CCl4	$K = 3$	4.0	2.00	15	-1706.6	3515.9	-113.4	69.9%	27.3%	2.7%	Constrained Model